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Models, Simulations, and Their Objects

1. Mathematical Models and Computer Simulations

Mathematical models and their cousins, computer simulations, occupy an uneasy space between theory and experiment, between abstract and concrete, and often between the pressures of pure science and the needs of pragmatic action. The contributions to this volume explore those uneasy spaces, and the work that it takes to maintain positions in those spaces.

Models and simulations do not, of course, form a homogeneous category. The ones considered here form a continuum, from spare symbolic entities to somewhat more complex sets of equations that are computerized largely for ease of calculation and manipulation, to computer programs so large and intricate that no one person understands how they function. The differences between the endpoints of this continuum are large enough that complex computer simulations can be said to use models, of many different types, or to have some particular models at their heart. Simple models and complex simulations, then, are in at least this way different types of objects, while they are related as endpoints on a continuum. Nonetheless, in being seen as occupying a position between theories and data, simulations and models perform some similar functions, and pose some similar problems. Although there are two sections in this volume, the first mostly addressing simulations and the second looking at economic models, some of the lessons of the papers cut across this divide of subject matter, applying to models and simulations, and to economics, physics, and physiology.

Whereas theories, like local claims, can be true or false, models and simulations are typically seen in more pragmatic terms, being more or less useful, rather than more or less true. Scientific models and simulations are given the status of tools, as well as representations; they are objects, as well as ideas. They easily cross categories, such as "theory" and "experiment," the bounds of which are otherwise well-established. And modeling and simulation sit uncomfortably in science both socially and epistemically, because of the boundaries they cross.

Models have become ubiquitous in public policy and corporate strategy, as well as applied and pure science. The demands of objectivity in public life have meant that decisions across a wide range of subject matters have to be accompanied by the appropriate scientific validation. Despite the fact that they do not have

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well-established epistemic positions within pure science, models and simulations form bridges between theoretical knowledge and its application to phenomena, bridges on which validation can be made to run.

Nelson Goodman points out that the word "model" is promiscuous in its meanings (Goodman 1968, 171; see Winsberg, this volume). Models are typically *copies* that manage to be both concrete examples and exemplary, in differing proportions. *Scale models* might be the simplest of models, being material copies of real or imagined (as in a model of an unbuilt architectural project) objects; they become epistemic instruments, and thus exemplary, if used or manipulated. The model student is one who stands between an ideal of studenthood and the bulk of students, examples of the one and to the others. Even the fashion model gives shape to clothing, but that shape is commonly thought of as ideal. An artist's model provides a concrete example — a copy of ideal poses, shapes, or scenes? — for the artist to represent.

Scientific uses of the term are no less promiscuous, and often sit in the same space of being both representations and things to be represented. The basic and original scientific models are material and conceptual analogues. These are manageable systems, or systems thought to be comprehensible, that stand in for unruly or opaque ones. For the turn-of-the-century physicist and meteorologist C. T. R. Wilson, his cloud chamber was a model system for studying cloud formation around ionized particles (Galison 1997). Tinker-toy models are material objects that can stand in for molecules, allowing chemists to manipulate and visualize things they cannot see; Eric Francoeur (1997) describes the genesis of these models, and the compromises that were struck between their requirements as flexible material objects and their requirements as accurate and persuasive depictions. Similarly, cell cultures are model systems for biochemistry, and rats and mice are model organisms for experiments on behavior and physiology, allowing for the production of knowledge about structures of behavior and physiology of a much broader class of organisms (Rheinberger 1997, Sismondo 1997). And conceptual models, like the billiard ball model of gases, posit analogies to make theories more comprehensible, to allow theories to be extended, and to give them explanatory force (Hesse 1966).

Mathematical models, the models of this volume, are similarly manageable systems standing in for the unruly or opaque, though also for the incomplete: they are typically seen as applications, approximations, or specifications of theories and principles that cannot by themselves be applied. One simple model for fisheries management, for example, is summed up by this central equation:

$$B_{t+1} = B_t + rB_t\,(1 - B_t/K) - qE_tB_t$$

where B_t is the biomass of fish in year t, r is the growth rate, K is the equilibrium size of the population without fishing, q is a coefficient representing the ease of catching fish, and E_t is the fishing effort during year t (Hilborn and Mangel 1997, 242–3). The model makes use of a basic understanding or theory of the dynamics

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of populations, but also of estimates of the fecundity of its organisms, the carrying capacity of the environment, and so on — it could easily be expanded to give a more complex representation of any of these factors, and to add others — to create an expected trajectory of population size. Models, then, have theories as inputs, and in so doing connect theories to data; they generally make more precise predictions than do theories, or make predictions where the theories can make none.

The material uses of the word "model" suggest that even mathematical models should be analogues of physical systems, and as analogues they are tools for understanding, describing, and exploring those systems. They should *behave* in the same way as the things they represent behave. Models are therefore different from theories not only in being applied, but in being analogues. Theories are now typically thought not to represent particular physical systems or even classes of physical systems, but underlying structures or necessities. According to the "semantic" conception, the current favorite philosophical account of theories, theories are defined by or specify families of models; theories provide structural ingredients of models. A distinction between models and theories is not always made in practice, however; and even when a distinction is drawn, some things are called models by people who want to call attention to their inadequacies, and theories by people who want to call attention to the insight that they provide.

Mary Hesse chooses as one origin point for debates about scientific models the comments and arguments of Pierre Duhem in his 1914 book *La Théorie Physique*. There Duhem ties models to national characters:

> This whole theory of electrostatics constitutes a group of abstract ideas and general propositions, formulated in the clear and precise language of geometry and algebra, and connected with one another by the rules of strict logic. This whole fully satisfies the reason for a French physicist and his taste for clarity, simplicity and order. ...
>
> Here is a book [by Oliver Lodge] intended to expound the modern theories of electricity and to expound a new theory. In it are nothing but strings which move around pulleys, which roll around drums, which go through pearl beads ... toothed wheels which are geared to one another and engage hooks. We thought we were entering the tranquil and neatly ordered abode of reason, but we find ourselves in a factory. (Quoted in Hesse 1966, 2)

While many English scientists would not have seen the factory metaphor as an insult — William Ashworth (1996) shows the impact of descriptions of the brain as a factory for thinking on Charles Babbage's ideas for his Difference Engine — Duhem pushes what he sees as an insult further, making the well-known contrast between the "strong and narrow" minds of Continental physicists, and the "broad and weak" ones of the English. Duhem is keen not only to support French over English science, but to support a picture of physical theory, soon to become roughly the Logical Positivists' picture, as the creation of abstract and elegant

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structures that predict and save empirical data. Hesse, drawing on the response to Duhem by the English physicist N. R. Campbell, sees models as heuristically essential to the development of and extension of theories, and also essential to the explanatory power of theories. Being able to derive statements of data from theories is not to explain that data, though being able to derive statements of data from theories via intuitive models is. Hence successful theories need models that concretize, that provide metaphors in which to think. Hesse's 1960s work on models and metaphors can now be seen as one of the arguments for the semantic conception of theories, as opposed to the "received" view, no longer so received:

The theoretical models that Hesse describes are first and foremost analogies, ways of bringing a theory to bear on the world by seeing one domain in terms of another. The wave theory of light creates an analogy between light and waves in media like water or air. But scientific models are much more various than that, many of them depending upon no analogy between domains, unless one of those domains is the mathematical formalism of the model itself, a possibility that strains Hesse's picture beyond recognition. For example, the fisheries model above doesn't posit any analogy (though there are undoubtedly many metaphors hidden in the background), but itself stands in for different factors affecting populations. The model does not essentially make use of some other analogy, but is itself something of an analogue. By being an analogue it is expected to have outputs that can be compared with data.

Computer simulations are similarly analogues, virtual copies of systems. Because of the problems that computers are seen to solve efficiently, simulations are usually more obvious analogues than are models. In most social simulations, for example, the researcher defines "agents" with fixed repertoires of behaviors. Those agents are then set to interact, responding at each moment to the actions of others. The goal is to see what, if any, social patterns emerge, or what assumptions are needed to create particular social patterns. Social simulations are analogues, then, because components of the program are analogous to individuals, and those components interact in a timeframe that is analogous to a sequence of moments in real time. Simple one-to-one correspondences between virtual objects and real ones (however idealized and simplified they might be) can be easily drawn. For most non-computerized mathematical models such correspondences are more difficult to draw, because preferred mathematical styles occlude individuals in favor of aggregates or larger-scale relations. The components of an equation can only rarely be neatly paired with obvious objects in the world, instead following a logic defined by relations among objects.

2. The Truth of Models and Theories

Because of their applicability we might want to say that a good model or simulation is more true than theories to which it is related. It makes predictions that are more

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precise or more correct than associated theories, if those theories make any predictions at all; for this reason Nancy Cartwright sees good models in physics as correcting theories and laws. Cartwright's work turns on recognizing the importance of *ceteris paribus* clauses. Most explanations involving fundamental physical laws use implicit *ceteris paribus* clauses, either to account for the fact that the fundamental laws assume an idealized world, or to account for complex interactions among causes. To take an example from her *How the Laws of Physics Lie* (1983) even so simple a law as Snell's law for refraction needs to be corrected.

> *Snell's Law*: At the interface between dielectric media, there is (also) a refracted ray in the second medium, lying in the plane of incidence, making an angle q_t with the normal, and obeying Snell's law:
>
> $$\sin q / \sin q_t = n2/n1$$
>
> where $v1$ and $v2$ are the velocities of propagation in the two media, and $n1 = (c/v1), n2 = (c/v2)$ are the indices of refraction. (Miles V. Klein, in Cartwright 1983, 46)

But, as Cartwright points out, this law is for media which are isotropic, having the same optical properties in all directions. Since most media are optically anisotropic, Snell's law is generally false, and needs to be corrected if it is to be applied in particular situations.

While fundamental laws and theories lie about real situations, models that use them or apply them can often get much closer to the truth. To see an easy case of that, we might turn to a model of climate change. The basic theory of the "greenhouse effect" is simple and firmly established. At equilibrium, the energy absorbed by the earth's surface from the sun would balance the energy radiated from the earth into space. However, the earth's surface radiates energy at longer wavelengths than it absorbs, which means that accumulations in the atmosphere of certain gases, like carbon dioxide and methane, will change the equilibrium temperature by absorbing more energy at longer wavelengths than shorter ones. General Circulation Models, which are computer simulations, address questions about the effects of greenhouse gases by assessing the interaction of many factors, including: water vapor feedback with temperature, the effects of clouds at different heights, the feedback with temperature of ice and snow's reflection of light, the changing properties of soil and vegetation with temperature, and possibly the effects of ocean circulation (IPCC 1995). Without taking into account these interacting factors, and more, the basic theory says almost nothing useful, or even true, about the real material effects of greenhouse gases.

Scientific theories are typically about idealized or abstract realms, which are uncovered in the process of developing and establishing the theories. Because theoretical reasoning is highly mobile, theories can achieve universality of recognition and respect that more particularistic scientific knowledge is less likely to attain. In addition, theoretical reasoning often does important work in explaining particulars; explanation usually is the placing of particulars in a persuasive theo-

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retical context, often one which indicates key causes. Therefore the realms that theories describe can be taken as fundamental and real, despite their lack of immediacy.

Some vaguely parallel points might be made about experiments. The category of "experiment" has been the subject of sustained inquiry in Science and Technology Studies for the past twenty years, and in that time it has been shown to be considerably more interesting, both epistemologically and socially, than it appears at face value (e.g. Hacking 1983; Latour 1983; Knorr Cetina 1981; Shapin 1984; Rheinberger 1997). With the exception of natural experiments, in which scientists achieve material control by drawing close but often problematic analogies between different naturally occurring situations, experiments are about induced relations between objects that are themselves often pre-constructed and purified. As such, experiments' epistemic value depends upon the careful social discrimination between the natural and the artificial or artifactual. Science's line of discrimination between natural and artificial became roughly fixed toward the end of the seventeenth century, and its exact location is renegotiated in different sciences on an ongoing basis.

For solid theoretical and experimental claims, then, the object of representation is given by nature, but it has become transparent with the acceptance of theoretical and experimental styles of reasoning. In both cases the object domain is only made manifest by human agency: theoretical claims are about idealized, transcendent, or at least submerged structures, the details of which become clear with theoretical work; experimental claims are about material structures which are almost always not found in nature, but are rather constructed in the laboratory. And each of theory and experimentation are given more epistemological value because of correspondences that are made between individual theories and experiments: strategies of correspondence are well-developed.

Some of those strategies involve models and simulations, which form bridges between theory and data. Gaston Bachelard argued that science alternates between rationalism and realism. In its rationalist mode it creates complex mathematical theories. In its realist mode it produces objects to match those theories, experimentally. This picture is a good starting point for at least some sciences, but should be supplemented by some understanding of the gap between theories and even experimentally produced reality. Models help fill that gap, and thereby legitimize the continuation of work on theory.

Mathematical models and computer simulations apply more concretely than the theories they instantiate, but because they don't have the epistemic traditions behind them that theorizing and experimentation have, they don't have transparent object domains. Models and simulations are typically considered neither true nor false, but rather more or less useful. Whereas the agency embedded in theoretical and experimental knowledge *per se* has become largely hidden, models and simulations are complex enough that they cannot be seen as natural: it is easy to see the assumptions made in modeling. This is despite the fact that, following

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Cartwright's down-to-earth intuitions, they are often more true to the material world, and thus more natural, than at least the theories to which they are related. Those theories, after all, make more assumptions about the material world, not fewer. But the agency behind models and simulations is too visible to allow them to easily represent transparent domains. They don't have a "home" domain, a metaphysical category to which they unproblematically apply. Unlike heavily idealized theories, they don't apply neatly to rationally-discovered Platonic worlds. Unlike local facts, they don't apply neatly to natural or experimental phenomena. One result of this is that when we see people arguing over a model, they are likely also arguing over styles of inquiry and explanation. Sismondo (1999) argues that a debate over an ecological model was in part an argument over what sort of a knowledge ecology would allow, whether it would allow highly abstract theoretical knowledge or only knowledge very tied to the material world. Breslau (1997) argues that a dispute in the U.S. over the evaluation of federal training programs is a dispute over the merits of different approaches to socio-economic research.

In the end, the above sort of talk of the domains of theory, experiment, and models takes us back to an old problem. How should we understand the status of object domains that appear to depend upon human agency, but have the solidity that we attribute to independently real structures? We can explain the solidity in terms of discipline, and assume that all such object domains are constructed. We can explain the discipline in terms of solidity, and assume that human agency wraps itself around the real. Or we can try to find terms in which to combine or evade constructivist and realist discourses. Although this is not a question directly addressed by any of the studies of this volume, taken as a whole they suggest that some version of the last option is the only option: models and simulations are misfits that don't sit comfortably in established categories.

3. The Richness of Modeling and Simulation

Michael Redhead (1980) distinguishes two types of theoretical models in physics. In cases in which a theory is difficult to apply, models are used which simplify assumptions or substitute tractable equations for intractable ones; these models are *impoverished* theories. Conversely, theories may provide constraints but leave space for more complete specification; then models are introduced which *enrich* the theory by filling the empty spaces. In either case, the ostensible goal of modeling is to apply theories, to connect the theories to data. In practice such models may be quite far from any conceivable data, being created in contexts in which some supplement to theoretical constructs is desired, to bring those constructs in *closer*, but not necessarily immediate, contact with some more actual or actualizable world.

Redhead, however, is perhaps looking at special cases. Most models and simulations have a more nuanced and distanced relation to theory, not merely impover-

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ishing or enriching theories. Even in their relation only to theory, simulations and complex models quite often violate Redhead's dichotomy by belonging to both sides. Simulations may be enriched theories, adding considerable detail in the effort to portray their objects, and simultaneously impoverished, substituting approximations for intractable equations, in the effort to make them more applicable. Both movements away from theory, though, are driven by the goal of applicability: theoretical structures which are computationally unmanageable, or unmanageably abstract, are inapplicable. Simulations, like models, stand between theories and material objects, pointing in both directions. For example, a simulation of AIDS transmission in intravenous drug users in a single country might have parameters that represent quite detailed knowledge of drug use, the population structure of drug users, responsiveness to treatment, and so on. At the same time it may incorporate simplifying assumptions, relative to other models, about the closure of its population, about the incubation pattern of AIDS, and so on (see, e.g., Pasqualucci et al. 1998).

In addition, putting models and simulations merely in the context of theories misses their often complex positioning in the world. Sometimes a theory will be put to work in applied science, but if so it is combined with so many other types of knowledge that it becomes only one component of a model or simulation. Take, for example, Ragnar Frisch's 1933 "Rocking Horse Model" of the business cycle, discussed in an elegant paper by Marcel Boumans (1999). The model consists of three equations, relating such quantities as consumption, production, and the money supply. As Boumans explains, there is a motivating metaphor behind the model; Frisch imagines the economy as a "rocking horse hit by a club." The model then brings physical knowledge to bear on the problem, through an equation normally used to describe a pendulum damped by friction. Yet Frisch was not merely enriching (or impoverishing) physical theory, because there are too many other components of the model: basic economic relations, a theory about delays in the production of goods and the deployment of capital, guesses of values of key parameters, and so on. As Boumans argues, economic model-building is the "integration of ingredients" so that the model meets "criteria of quality." Theories play a role, but so do metaphors, mathematical techniques, views on policy, and empirical data; for example, Frisch chose values for parameters to make sure that his model would cycle at the same rate as real economic cycles. In addition, there may be multiple theories playing roles, from different disciplines. Therefore we shouldn't see models and simulations only in their relation to theories, bringing theories into closer contact with data; models and simulations do many things at once.

Seeing models and simulations just in a space between theories and data, the typical way of seeing them, misses their articulation with other goals, resources, and constraints. There are resources provided and constraints imposed by the media in which they operate, because they are created with the mathematical and computing tools available. They are created to fit into particular cultures of

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research: models and simulations have to take particular forms in order to be accepted. And they are created to fit into particular social settings, becoming objective by balancing among sides in debates.

4. Modeling and Simulating as Experiment and Theory

Work on models and simulations is like theoretical work in that the ostensible object of representation is absent; modelers and simulators trade in symbols. They produce representations, perhaps check those representations against data, and so on. But modeling and (especially) simulation is like experimental work in that the behavior of the model or simulation is the subject of investigation. From her many interviews with people engaged with simulations, Deborah Dowling shows that working with simulations is seen to have aspects of experimental work, despite its being largely in the realm of representation. Researchers make small changes — to parameters, initial conditions, the grain of calculation, etc. — and learn what results. The flavor of such activities better matches the flavor of experimentation than that of theorizing. Thus Dowling describes the ways in which simulations are pulled back and forth between the categories of theory and experiment, depending upon context.

Modeling and simulation is also like experimentation in its pattern of give and take in their creation. Daniel Breslau and Yuval Yonay frame their contribution to this volume within a critique of the singular focus on metaphor in studies of economics. Illustrating a microinteractionist approach to the sociology of economics, they draw on studies by David Gooding, Ludwig Fleck, and Andrew Pickering, to show that economic modelers have to perform, in Pickering's term, a "dance of agencies" in much the same way as do experimenters. The materials that modelers work with — particular established formulations and modeling tools — are recalcitrant; they do not behave as the modelers would like, either producing too many answers, the wrong sort of answer, or intractable equations. In response, the modelers have to find assumptions and tools that allow them to create objects with the right disciplinary forms, objects capable of indicating unique solutions to problems, sophisticated enough to be seen as advances in the field, and uncontrived enough to produce epistemic solidity. This last condition is particularly important: to avoid too-easy criticisms of assumptions, those assumptions have to conform to disciplinary standards. As a result, the model is granted agency by the modelers.

Epistemic solidity for a model or simulation is tricky. The criteria that might be applied depend upon the uses to which the object might be put, its complexity, the available data, and the state of the field. As Eric Winsberg shows in his paper in this volume, simulations and their components are evaluated on a variety of fronts, revolving around *fidelity* to either theory or material: assumptions are evaluated as close enough to the truth, or unimportant enough not to to mislead; approximations are judged as not introducing too much error; the computing

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tools are judged for their transparency; graphics systems and techniques are expected to show salient properties and relationships. All of these judgments and more are difficult matters, and many do not have straightforward answers. In some cases these judgments create an interesting type of uncertainty. In the global warming debate, for example, a key type of number has become the estimate of the sensitivity of temperatures to a doubling of carbon dioxide in the atmosphere. Sensitivity is typically reported as a range, as in 1.5° to 4.5°, or as an estimate with upper and lower bounds, as in 3° ± 1.5°. The range does not measure statistical uncertainty, though, because the key General Circulation Models are made to be deterministic. Rather, the range measures the *confidence* of the climatologists in their models and their assumptions (Sluijs et al. 1998).

In their focus on examining the warrant for fundamental theories, philosophers of science have almost completely neglected the processes involved in applying such theories. When philosophers do address application, it is generally assumed that application is little more than deriving data from equations, with the help of the right parameters and assumptions. But the papers of this volume show that modeling and simulation, typical modes of application, are anything but straightforward derivation. Applied theory isn't simply theory applied, because it instantiates theoretical frameworks using a logic that stands outside of those frameworks. Thus Winsberg calls for an "epistemology of simulation," which would study the grounds for believing the results of complex models and simulations.

5. Models and Simulations as Tools and Objects of Knowledge

Because they are supposed to be analogues, models and simulations are themselves studied in the way that natural systems might be. Knowledge about them is supposed to run parallel to knowledge about the things that they represent, which allows modeling to be like experimentation, in both Dowling's and Breslau and Yonay's senses. Researchers can learn about the behavior of models and simulations, or make them behave naturally, and be doing properly scientific research. But as objects they are also open to use in more instrumental contexts, providing inputs for other research. That is, if treated as black boxes the data they produce can sometimes simply feed into other research.

Martina Merz's paper here is an ethnographic study of event generators, computer programs in high energy physics that simulate the effect of beams of particles colliding with other particles. They are important for a number of reasons: they play a role in data analysis, being used as a template for comparison with real data; they are used to test simulations of detectors. But they are not just tools, because creating them and using them is doing physics, too. Merz argues, using terminology adopted from Hans-Jorg Rheinberger and Karin Knorr Cetina, that event generators manage to be simultaneously epistemic objects and technological things. As epistemic objects what is valued is their open-endedness, their ability to behave

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unpredictably when pushed. As technological things what is valued is their closure, their assumed straightforward application of taken-for-granted knowledge. In Merz's rich treatment of the use and study of event generators she shows how this divide between epistemic objects and technological things plays out in the culture of particle physics. Actors' differing fields of interest mean that event generators are expected to remain both open and closed, and so are slated to remain multivalent objects.

Interestingly, the same divide can be seen in the relatively simple economic and econometric models that Robert Evans and Adrienne van den Bogaard discuss in the second section of this volume. For some actors they are epistemic things, or as Rheinberger calls them, "question-generating machines." In the attempt to make them policy-relevant their proponents try to turn these models into technological objects or "answering machines." For Evans, democratic participation in economic decision-making requires processes that keep alive the multivalent appearance of economic models.

6. Models and Simulations Negotiating Politics

Models and simulations are, as I have already mentioned, increasingly important in more public spheres. The two last papers of this volume take a look at the mediating work of models in economic policy, and the boundary-crossings they describe leave the sphere of relatively pure research.

Robert Evans examines a controversy over economic models to show how the models concretize assumptions that have obvious moral weight, in his case assumptions over the causes of unemployment. The models in question were used by the different members of the United Kingdom's "Panel of Independent Forecasters," which from 1993 to 1997 was a key actor providing information for the UK's economic policies. As Evans shows, there is great uncertainty surrounding the models: it isn't clear which ones make the best predictions, and which ones have the firmest evidential base. Hence it is possible to see a dispute about morals which is integral to the dispute about models. While unique solutions to central problems of economic modeling would undoubtedly have constrained economic policy choices, the inability of the modelers to agree on unique solutions created a situation in which the values implicit in policy decisions could be sharply defined and highlighted.

Adrienne van den Bogaard presents an overview of her research on the development of the institution of economic forecasting in the Netherlands in the 1930s and 1940s. She argues that a particular style of macro-economic and macro-econometric modeling of the Dutch economy became central to economic planning because it solved political problems. In particular, it solved problems of objectivity, by not being identifiable with the perspectives of any of the four traditional "pillars" of Dutch society: the social democrats, the liberals, the catholics, and the

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protestants. Each of these pillars came with its own politics, down to the level of economic decisions. At a time when fractures were severe, economists had to find an authoritative approach. A failed approach attempted to synthesize the perspectives of the pillars. The successful approach turned on a confluence between the neutrality of modeling and the particular style of Dutch tolerance, a careful and studied tolerance that recognizes the need to grant factions and positions their autonomy, even while not granting them respect. The successful model could become objective by being neutral, thereby becoming a tool for and an arbiter between all of the pillars; the successful modelers could create a monopoly over some central portions of economic planning.

In short, models and simulations cut across boundaries of pure categories we accept in science, and sometimes politics. Some people might be tempted to see the compromises that models make — between the domains of the theoretical and the material, between their uses as pragmatic and representational objects, between different goals — as unsatisfactory, to see it as simple inconsistency or imperfection. But we might choose instead to see models and simulations as monsters necessary to mediate between worlds that cannot stand on their own, or that are unmanageable. The level of the ideal, for example, often lacks legitimacy among the instrumentally-minded. The natural world is usually intractable in terms of ideals, but it is opaque without them. Models become a form of glue, simultaneously epistemic and social, that allows inquiry to go forward, by connecting the ideal and the material. To do that they need to make compromises: they must simultaneously look like theory — because they have to explain, predict, and give structure — and like practical knowledge — because they have to connect to real-world features. They are a diverse lot, and are made to do a diverse number of things. But they are necessarily so, being made to stand between worlds, and pushed one way and another. Therefore we should resist the urge to do much epistemic neatening of this messy category of models and simulations.

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References

Ashworth, William. 1996. "Memory, Efficiency, and Symbolic Analysis: Charles Babbage, John Herschel, and the Industrial Mind." *Isis* 87:629–653.

Breslau, Daniel. 1997. "Contract Shop Epistemology: Credibility and Problem Construction in Applied Social Science." *Social Studies of Science* 27:363–94.

Boumans, Marcel. 1999. "Built-In Justification." Forthcoming in M. Morgan and M. Morrison, editors, *Models as Mediators*. Cambridge: Cambridge University Press.

Cartwright, Nancy. 1983. *How the Laws of Physics Lie*. Oxford: Oxford University Press.

Francoeur, Eric. 1997. "The Forgotten Tool: The Design and Use of Molecular Models." *Social Studies of Science* 27:7–40.

Galison, Peter. 1997. *Image and Logic: A Material Culture of Microphysics*. Chicago: University of Chicago Press.

Goodman, Nelson. 1968. *Languages of Art*. New York: Bobbs-Merrill.

Hacking, Ian. 1983. *Representing and Intervening: Elementary Topics in the Philosophy of Natural Science*. Cambridge: Cambridge University Press.

Hesse, Mary. 1966. *Models and Analogies in Science*. Notre Dame: Notre Dame University Press.

Hilborn, Ray, and Marc Mangel. 1997. *The Ecological Detective: Confronting Models with Data*. Princeton, NJ: Princeton University Press.

IPCC (Intergovernmental Panel on Climate Change). 1995. *Climate Change 1995: The Science of Climate Change*, edited by J. T. Houghton, L. G. Meira Filho, B. A. Callander, N. Harris, A. Kattenberg, and K. Maskell. Cambridge: Cambridge University Press.

Knorr Cetina, Karin D. 1981. *The Manufacture of Knowledge: An Essay on the Constructivist and Contextual Nature of Science*. Oxford: Pergamon Press.

Latour, Bruno. 1983. "Give Me a Laboratory and I will Raise the World." In *Science Observed: Perspectives on the Social Study of Science*, edited by K. D. Knorr Cetina and Michael Mulkay. London: Sage.

Pasqualucci, Cristina, Lucille Ravà; Carla Rossi, and Giuseppe Schinaia. 1998. "Estimating the Size of the HIV/AIDS Epidemic: Complementary Use of the Empirical Bayesian Back-Calculation and the Mover-Stayer Model for Gathering the Largest Amount of Information." *Simulation* 71:213–227.

Redhead, Michael. 1980. "Models in Physics." *British Journal for the Philosophy of Science* 31:145–63.

Rheinberger, Hans-Jorg. 1997. *Toward a History of Epistemic Things: Synthesizing Proteins in the Test Tube*. Stanford: Stanford University Press.

Shapin, Steven. 1984. "Pump and Circumstance: Robert Boyle's Literary Technology." *Social Studies of Science* 14:481–520.

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Sismondo, Sergio. 1997. "Deflationary Metaphysics and the Construction of Laboratory Mice." *Metaphilosophy* 28:219–32.

——. 1999. "Island Biogeography and the Multiple Domains of Models." *Biology and Philosophy*, forthcoming.

Sluijs, Jeroen van der, Josée van Eijndhoven, Simon Shackley and Brian Wynne. 1998. "Anchoring Devices in Science for Policy: The Case of Consensus around Climate Sensitivity." *Social Studies of Science* 28:291–323.

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